**Post Replied To**
by Dzemal Jukic - Monday, 3 August 2026, 9:46 PM
Number of replies: 2

Industry 4.0 technologies, including automation, cyber-physical systems, Internet of Things devices and interconnected information systems have significantly transformed the logistics sector (Efthymiou and Ponis, 2021). They allow logistics companies to obtain real-time information, improve the visibility and monitoring of goods, and coordinate activities across warehouses, transport networks and supply chains (Chauhan et al., 2023). Industry 4.0 systems also generate and process large volumes of operational data, supporting faster decision-making and more efficient industrial operations (Raptis, Passarella and Conti, 2019). However, increased connectivity can also create vulnerabilities because a failure affecting an important information system may rapidly disrupt physical operations and interconnected supply chains. Cheung, Bell and Bhattacharjya (2021) observe that although internet-based technologies improve logistics services, their adoption also increases the sector's exposure to cyberattacks.

A significant example is the NotPetya cyberattack experienced by A.P. Moller–Maersk on 27 June 2017. The attack disrupted information systems across the company's transport and logistics operations, causing operational difficulties, reduced shipping volumes and disruption to customers (A.P. Moller–Maersk, 2018). During the recovery, Maersk reportedly had to rebuild approximately 4,000 servers and 45,000 computers within ten days (Greenberg, 2018). Employees also relied on improvised and partly manual processes while the company's digital infrastructure was being restored. The scale of this reconstruction demonstrates Maersk's dependence on globally interconnected information systems.

Maersk estimated that the attack reduced its profitability by approximately US$250–300 million through lost revenue, IT restoration costs and extraordinary operational expenses (A.P. Moller–Maersk, 2018). The impact also extended beyond the company itself. Crosignani, Macchiavelli and Silva (2023) found that the effects of NotPetya spread from directly affected organisations to their customers, producing a fourfold amplification of the initial reduction in profits. This demonstrates how the failure of a major logistics provider's information systems can affect organisations throughout a supply chain.

The incident highlights an important limitation of Industry 4.0: systems designed primarily for integration, connectivity and efficiency may become vulnerable when the same connectivity allows failures to spread across systems and locations. Metcalf (2024) explains that Industry 5.0 initially emerged partly as a reaction to the excesses associated with Industry 4.0 technologies. From an Industry 5.0 perspective, logistics systems should therefore prioritise resilience and human involvement alongside automation and efficiency. Measures such as network segmentation, offline backups, tested recovery procedures and manual operational alternatives could reduce the effects of similar incidents. The Maersk case demonstrates that human adaptability remains essential, but it should complement resilient system design rather than compensate for its absence.

## References

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**My Response**

Dear Dzemal,

Your post suggests that logistics systems have evolved to the point where they're especially vulnerable to cyber attacks and general system failures and that this is a limitation of Industry 4.0.

Metcalf (2024) agrees with this point of view and states that Industry 5.0 is the phase of attempting to balance out this "excessive reliance on technologies". I agree that the reliance of many industries, not just logistics, on modern-day connectivity and instantaneous information is now a weakness that is exposed and needs to be guarded against.
This need to develop resilience and flexibility within industry is also highlighted in a European Commission report (2021), which speaks about the requirement for resilience in terms of technology, but also goes further and addresses the requirement for sustainability to ensure that not only are our technology systems sustainable, but the resources we use in order to support this are re-usable where possible so as not to unnecessarily deplete our planet's resources.

The report suggests that the drive towards sustainability and resilience should focus on ensuring that basic human needs can always be met, despite any interruption to technology, whilst minimising any effects that the support of technology may have on the planet's resources.

References

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